

The Onset Time of the Dynamic and Static Invisible Body Illusion

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ABSTRACT

The body part ownership is elicited in 23 s, and the full-body illusion is produced in 5 s. Our previous study has shown that synchronized hands and feet movements produce the full-body illusion at an empty space between the hands and feet in 5 min at most, but the exact onset time is still unclear. This study investigated the optimal learning method and learning time for the invisible body illusion by measuring the onset time of the illusion in active movement and static observation. As a result, the illusion was produced after 6.93 s of synchronized movement or 4.67 s of static observation. These results suggest that both active movement and static observation are effective for the invisible body illusion.

Keywords: Body ownership, full-body illusion, invisible body

Index Terms: I.3.7 [Computer graphics]: Three-Dimensional Graphics and Realism—Virtual Reality; H.1.2 [Models and Principles]: User/Machine Systems—human factors

1 INTRODUCTION

Body ownership (the feeling that the body belongs to me) is induced not only in our biological body but also in a fake body such as a rubber hand [1]. This is called the rubber hand illusion, in which a participant is stroked their hand and the rubber hand by a brush at the same time, and when the participant observes only the rubber hand, he/she feel the rubber hand as if it is his/her own hand [1]. In addition to the visual and tactile synchronous stimuli, the illusory ownership of body parts (body part ownership) is also elicited by visual-motor synchronization, in which a participant and a model hand move synchronously [2]. In the same way, the illusion of body ownership is generated not only for body parts but also for a full body (full-body illusion) [3], [4]. The rubber band illusion, which is the body part ownership, is produced in 23 s [5], and the full-body illusion in 5 s [6]. Interestingly, the full-body illusion is induced by observing the full-body avatar for only 5s [6].

Thus, although the onset time of the body ownership for the rubber hand and the full-body avatar has been reported, the onset time of the body ownership in the invisible body [7] has not been clarified. Our previous study [7] has shown that synchronized hands and feet movements produce the full-body illusion at an empty space between the hands and feet in 5 min at most, but the exact onset time is still unclear. This study aimed to investigate the optimal learning method and learning time for the invisible body illusion by measuring the onset time of the invisible body illusion in the moving and stationary states.

2 METHOD

Ten volunteers participated in the experiment. They had a healthy vision and motor abilities. The study was conducted with the approval of the Ethical Committee on Human-subject studies of Toyohashi University of Technology.

Participants grasped two Valve Index controllers with their left and right hands and wore shoes with HTC Vive Trackers in order to capture the hand and foot movements. A pair of gloves holding controllers and a pair of socks were presented in the same position as the participant's hands and feet in a VR space created with a computer (OS: Windows 10 (64bit), CPU: Intel Core i7-9700F, RAM: 16GB, GPU: NVIDIA GeForce RTX 2070 SUPER 8GB), and the participant observed them from the first-person perspective through a head-mounted display (HMD: Valve Index, 2880 x 1600 pixel, 90Hz, 130deg). A mirror reflecting the gloves and socks was placed 1m in front of the participant. There were three conditions: synchronous condition in which the gloves and socks moved in synchrony with the participant's hands and feet (Fig. 1, left), asynchronous condition in which the gloves and socks moved with a 1.75 s delay, and static condition in which the participants looked down and did not move, and the gloves and socks did not move (Fig. 1, right). In all conditions, a pure-tone (1 kHz, 500 ms) was presented to both ears through the headphones of the HMD every second (1Hz) to synchronize the timing of the participant's movements.



Figure 1: Synchronous condition (left), Static condition (right)

In the synchronous and asynchronous conditions, the participants moved each body part in the order of the right hand, the left hand, the right foot, and the left foot for 60 s following the sound presented every second from the headphones. In the static condition, participants looked down and stayed there for 60 s. At the end of each condition, the participants answered 5 questions about body perception on the seven-point Likert scale (-3: did not feel at all, 0: neither, +3: felt very strongly). The order of each condition and the order of the questions were random.

Questionnaire

1. It felt as if the space between the gloves and socks was my body.
2. The movement of the gloves and socks seemed to be my movement.
3. It felt as if I had two bodies.
4. It felt as if the gloves and socks were controlling my movements.
5. It seemed as if there is a whole body between the gloves and socks.

After these tests were performed once in each condition, the participants whose rating of the item concerning the full-body illusion (Q1) was +1 or higher in the synchronous condition and -1

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or lower in the asynchronous condition were measured the onset time of the full-body illusion. The participants moved their bodies as in the synchronous condition, and the gloves and socks moved synchronously. They were instructed to press the button on the right-hand controller as soon as possible when they felt that the space between the gloves and the socks was their own body. After pressing the button, the participants continued the movement until the end of the trial, and this task was repeated three times. Among the participants who performed the time measurement task, participants who scored +1 or higher on the Q1 in the static condition were measured the onset time three times in the static condition. In order to eliminate the full-body illusion in the previous trials, the participants removed the HMD and moved their bodies freely for approximately 30 s between trials.

3 RESULTS

For each questionnaire item, multiple comparisons were conducted using Wilcoxon signed-rank test, and p-values were corrected by the Bonferroni method.

As a result, when the gloves and socks were synchronized with the participants' hands and feet, the space between the gloves and socks felt more like their own body than when the gloves and socks were asynchronous (Figure 2 Q1: $z = 2.83$, $p = .0059$, $r = .89$). When the participants, gloves, and socks were stationary, the space between the glove and sock felt more like my body than when they were asynchronous (Q1: $z = -2.70$, $p = .012$, $r = .85$). Interestingly, there was no significant difference in full-body illusion between the synchronous condition and static condition (Q1: $z = .30$, $p = 1.00$, $r = .10$). When the movement of the gloves and socks was synchronized with the participants' movement, the movement of the gloves and socks was felt more like their own movement than when it was asynchronous or stationary (Q2 Synchronous vs. Asynchronous: $z = 2.83$, $p = .0059$, $r = .89$; Synchronous vs. Static: $z = 2.68$, $p = .012$, $r = .85$). There was no significant difference in Q2 between when the gloves and socks were asynchronous to the participants' movement and when they were stationary (Q2: $z = -1.78$, $p = .28$, $r = .56$). Q3 and Q4 were control questions, and there were no significant differences between conditions. In Q5, I predicted that participants would see the invisible body only in the synchronous condition, as in our previous invisible body study [7], but there was no significant difference between conditions (Q5: Synchronous vs. Asynchronous: $z = 1.13$, $p = .28$, $r = .36$; Synchronous vs. Static: $z = 2.06$, $p = .04$, $r = .65$; Asynchronous vs. Static: $z = 1.26$, $p = .21$, $r = .40$).

Seven of the ten participants were measured the onset time of the full-body illusion in the synchronized condition. Four of the seven participants were measured the onset time in the static condition. The mean onset time was 6.93 s (SE ± 2.65 s) in the synchronized condition and 4.67 s (SE ± 1.01 s) in the static condition (Fig. 2). Although the onset time was faster in the static condition than in the synchronous condition, there was no significant difference between conditions (Wilcoxon rank-sum test: $z = 0$, $p = 1.00$, $r = 0$).

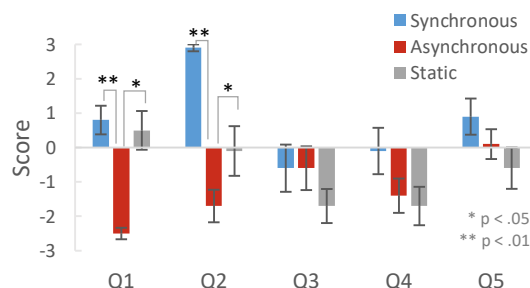


Figure 2: Results of questionnaire

4 DISCUSSION

In this study, the same intensity of full-body illusion was elicited in the synchronous and static conditions. There was no significant difference in the onset time of the full-body illusion between the two conditions. This suggests that both active movement and stationary observation are effective for learning to induce the invisible body illusion, and there is no difference in the learning time. In the stationary condition, the gloves and socks were synchronized with the participants' hands and feet in order to adjust the position of the gloves and socks, so that small movements may have triggered the full-body illusion.

The onset time in this study was comparable to the study in which full-body illusion was induced in 5 s [6], suggesting that the visual information of the body other than hands and feet does not affect the onset time of the full-body illusion. However, this study did not directly compare the full body with the invisible body. In further comparison, the relationship between the body presentation area and the onset time will be clarified.

ACKNOWLEDGMENTS

This study was supported by JST ERATO Grant Number JPMJER1701 and JSPS KAKENHI Grant Number JP19J12660, Japan.

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